Smart Irrigation System

Student: Biet Puoric Matuong | **Supervisor:** Eng. Lodu B. John Kuya

Problem & Motivation

- **Problem Context:** Inefficiencies and structural challenges of traditional agricultural irrigation in South Sudan, specifically observed at **Freedom Farm in Luri, Central Equatoria State**..
- **Research Gap:** Lack of **low-cost, solar-powered, and locally adaptable smart irrigation system** that automates water delivery using real-time soil moisture and environmental sensors tailored specifically to rural, resource-constrained farming environments
- **Objective:** Design and build a low-cost, sensor-based, solar-powered smart irrigation prototype tailored for local conditions.

Significance & Conclusion

- **Contribution:** Developed an affordable, functional Internet of Things (IoT) prototype that combines soil moisture, temperature, and humidity sensors with automated solenoids/pumps controlled by a microcontroller (ESP32/Arduino).
- **Impact:** 30%–40% Reduction in Water Usage. By triggering irrigation only when real-time soil moisture dropped below the designated threshold, the system eliminated unnecessary over-watering
- **Future Work:** Integrate AI to improve performance and the use of low-power wide-area networks (LPWAN), to address connectivity challenges in remote areas.

System Implementation & Results